Skeleton-transformer augmentation for BST on ShuttleSet

Companion pose-anchor continuity. For ML Team.

TL;DR

Four augmentations, applied per-clip in the DataLoader's __getitem__, chosen on literature + project-specific reasoning:

Augmentation	Parameter	Applied to
Centreline flip (mirror across court centreline; left-right in broadcast frame)	p=0.5	pose + shuttle + court positions, *coupled*
Temporal speed jitter (stretch + resample)	scale ∈ [0.85, 1.15], p=0.5	pose + shuttle + court positions, *coupled*
Gaussian joint jitter	σ = 0.005 in court-normalised units, p=0.5	pose only
Random joint masking	10% of 34 joints per clip, p=0.5	pose only

Plus a sampling-side intervention: WeightedRandomSampler with per-class weights ∝ 1/√(class_count), applied during training but not evaluation.
Parameters are literature-anchored starting points, not empirically tuned on ShuttleSet. Ari's ablation pass against the baseline decides.
First-order finding for the reconciliation table: BST's training loop applies exactly one augmentation. RandomTranslation_batch (joints-only xy-shift, range +-0.3, p=0.3), instantiated at bst_train.py:243 and applied at lines 93 and 98. Ported verbatim from TemPose. The dataset-level RandomTranslation class (shuttleset_dataset.py:108) is effectively dead code, called with its return value discarded in Dataset_npy.__getitem__ (line 202, silent no-op bug), and explicitly disabled in Dataset_npy_collated (docstring, line 240). Ari's "no augmentation" framing is substantially correct. The one augmentation that is live violates Rule 1 of §3, because pose is translated but shuttle and court aren't, which is independent corroboration of the multi-modal coupling problem §3 describes.

Scope: in-distribution training-signal expansion for skeleton-transformer classification. Out of scope:
(a) amateur distribution shift (that's Phase 3, and needs wider parameter ranges than recommended here);
(b) Arch 1 X3D-S wrist-crop RGB augmentation (different regime, different literature, separate concern).
Both flagged in §7.

1. The problem

BST (Chang, arXiv:2502.21085, 2025) and its predecessor TemPose (Ibh et al. CVPRW 2023) both train without reported skeleton augmentation. In a dataset where Rush has 358 train clips and Net shot has ~4,000+, that's a real gap. The decision boundary for rare classes is being drawn from a handful of player × match combinations, and the transformer has no exposure to plausible-but-unseen variations of those strokes.
Ari's has hypothesised was that BST's heuristic choices (player filtering, court-relative normalisation, framing) were silently compensating for insufficient per-class data. That's testable in principle. The right way to test it is to add carefully-chosen augmentation and see whether rare-class F1 lifts more than head-class F1 drops. But the first step is picking augmentations that are (a) label-preserving on this task, (b) plausible as real variation, and (c) compatible with the multi-modal input stack.

2. The constraint

"This regime" is narrower than it looks. Constraints from the call and the pipeline:

- **Input is 2D, court-normalised, multi-modal.** Clip tensor is (F, 2, 17, 2), F frames, 2 players (Top / Bottom by court side), 17 COCO joints, (x, y) in court-relative coordinates. Shuttle trajectory is (F, 2). Court positions are (F, 2, 2). These three streams are **synchronised in time and space**, same court frame of reference, same frame indexing. Any augmentation that transforms one has to transform the others consistently, or the transformer is learning mismatched modalities.
- **Model is a transformer**, not a GCN. Most skeleton-aug literature is GCN-heavy (ST-GCN, 2s-AGCN, CTR-GCN); augmentations that rely on graph structure (joint permutation, graph topology perturbation) don't transfer cleanly. Augmentations that act on raw coordinates and frame indices do.
- **Coordinates are court-relative, not world-relative or camera-relative.** This kills some standard augmentations outright. 3D rotation around the vertical axis makes no sense when coordinates are already a top-down court projection. 2D rotation also breaks court geometry unless the rotation is small. The "view invariance" motivation that drives a lot of NTU-dataset augmentation literature does not apply here.
- **Long-tail is severe.** Rush (train=358) and Long service (train=211) are the thin end. Augmentation needs to help these without dominating the head classes. Applied uniformly, augmentation helps all classes by the same multiplier; that's fine, but not the same thing as targeted long-tail correction. The sampling-side intervention (§5) is how we get class-aware amplification without touching the augmentation transforms themselves.
- **Applied at training time, in the DataLoader.** No precomputed augmented copies on disk. No separate augmented training pass. Every epoch sees a fresh random transform of the same 24k-ish training clips.
- **Parameters come from skeleton-AR literature and biomechanical plausibility, not a ShuttleSet ablation.** Ari's runs produce F1 deltas.

3. Multi-modal coupling rules

Most skeleton-augmentation literature (JMDA (Xiang et al. 2024), Shap-Mix (Zhang et al. IJCAI 2024), the NTU RGB+D 60/120 ablations, the contrastive-learning families (SkeletonCLR, AimCLR, A2MC)) assumes single-modality input: one skeleton per clip. BST isn't that. BST

takes three synchronised streams (pose, shuttle, court positions), and what the model is actually learning is the relationship between them: where the shuttle is relative to the player's wrist at the strike frame, which half of the court the top player is defending, etc.
This means an augmentation that looks innocuous on pose alone can wreck a cross-attention pattern. These rules apply regardless of which augmentation families end up in the final set.

Rule 1: Spatial transforms apply to all three streams or none

Horizontal flip, temporal speed, translation, scale: if applied at all, they must be applied identically to pose, shuttle, and court positions. A flipped skeleton with an un-flipped shuttle trajectory teaches the model that strokes produce shuttles on the wrong side of the court. This is the biggest failure mode to avoid.

Rule 2: Use the court-centreline flip, not the net flip

Two possible flip axes in the court frame: **left-right (across the centreline)** and **top-bottom (across the net)**. The centreline flip is straightforward: swap x -> W - x on all three streams. Label unchanged, physics unchanged, joint-index swaps for bilateral joints (see Rule 3). The net flip is fiddlier: it swaps Top and Bottom player slots, which means the label's implied player identity flips, which means downstream player-metadata features (dominant hand, height) need to be swapped too. It's also not standard in the skeleton-AR literature.
The recommendation below assumes centreline flip only. Net flip goes in the appendix under "considered, needs extra care."

Rule 3: Bone-vector representations transform differently from joint-position representations

BST supports multiple pose styles (J, J+B, J2B, J+2B). Joint positions flip linearly (x -> W - x). Bone vectors (differences between joint pairs) also flip linearly, but with a crucial gotcha: any bone whose parent/child assignment encodes laterality (e.g. "left shoulder to left elbow") has its **joint index permutation** swap under a mirror, even though the coordinates transform identically. If the code treats joint indices as canonical (joint 9 = left wrist, joint 10 = right wrist in COCO17) then a horizontal flip needs both a coordinate flip *and* a left<->right joint-index swap for the bilateral pairs: (5,6), (7,8), (9,10), (11,12), (13,14), (15,16). The nose (0), eyes (1,2), ears (3,4) and the torso pairs swap similarly. Skip this index swap and the model learns that a right-handed smash looks like a mirrored left-handed smash with the wrong joint-index semantics. The joint-level positional embedding becomes noise under flipping.

Rule 4: Temporal transforms preserve cross-modal frame alignment

If the pose stream is resampled from F frames to F' frames (temporal speed jitter), the shuttle stream and the court-position stream must be resampled by the identical index mapping. Nearest-neighbour indexing is the safe default for shuttle (discrete events like hit frames can't be interpolated meaningfully); linear interpolation is fine for court positions (slow-moving) and for pose (smooth except at the strike). If the clip carries any frame-index metadata (hit-frame indices, strike markers), those indices must be updated under the resample mapping. Off-by-one here is catastrophic for downstream fusion or for any head that attends to a specific frame.

Rule 5: Additive noise does not transfer across modalities

Gaussian jitter on joint positions (simulating pose-estimation noise) makes sense. Gaussian jitter on the shuttle trajectory does **not**. The shuttle is a physical object whose trajectory is constrained by flight physics, and TrackNetV3's error characteristics are completely different from MMPose's. Gaussian jitter on court positions is also wrong: court projection noise is a function of homography quality, not a stationary Gaussian. So additive noise is pose-only, and not even uniform within the pose stream: wrist noise is larger than hip noise in the real MMPose error distribution (per the 03_feature_validation work). A single σ is an approximation but a defensible one.

Rule 6: Masking operates at the joint level in one stream at a time

Random joint masking (setting random joints to zero or a learnable masked-value per frame) is a per-stream operation. Masking a joint in pose does not require masking shuttle or court positions. This is the only augmentation that genuinely lives in one modality. Keep per-sample mask rates low (~10%) and sampled fresh per epoch.

Rule 6b: Masking a joint is not the same as dropping a frame

If the codebase treats (0, 0) coordinates as "missing," a masked joint will be confused with a MMPose dropout. Two options: (a) use a sentinel value outside the normalised [0, 1] court coords (e.g. -1) and ensure the model's pose encoder handles it, or (b) use the mean joint position over the clip as the mask value (common in self-supervised skeleton literature). The BST code already has some convention for missing joints, check it before picking a sentinel.

4. Augmentation families: analysis

4.1 Spatial geometric transforms

Candidates: rotation, scaling, shearing, translation, horizontal flip.
The general-purpose skeleton-AR literature (surveyed by Liu et al. 2025, 3D Skeleton-Based Action Recognition: A Review, arXiv:2506.00915; and the MDPI Electronics 13(4):747, 2024 augmentation-specific study) treats this as the underlying layer: small random rotations, scalings, shears, and translations applied to joint coordinates, motivated by view invariance. Standard parameter ranges from that literature are rotation +-15° to +-30°, scaling ×[0.9, 1.1], shear [-0.1, 0.1], translation a few percent of the coordinate range.
Most of these do not translate cleanly to this regime. Court-relative coordinates aren't viewpoint-dependent. They're already projected into a standard top-down frame. A rotation of +-15° around the court centre moves Top player's feet off the top baseline and into the stands. A shear deforms court geometry, which the model cross-attends to via court positions; the shuttle trajectory starts looking non-physical. Scaling is similarly problematic: it changes the apparent player size, but in court-relative coords "player size" is a proxy for "is the court projection broken", not something to randomise.
The one that survives the court-geometry filter is **horizontal flip across the court centreline** (Rule 2 above). It's label-preserving (smashes still look like smashes), physically plausible (the court and game are symmetric left-to-right), and it genuinely doubles the effective training set for laterality-sensitive patterns. Shap-Mix (Zhang et al. 2024) and most skeleton-contrastive-learning papers include spatial flip as a core augmentation with p=0.5.
Translation is a corner case and the live issue. A small per-clip shift of all three streams by a shared offset (e.g. +-0.02 in normalised coords) mimics small court-projection calibration. The BST codebase already applies this (see Section 6, RandomTranslation). The decision is whether to keep, extend, or tune it. No clean literature-backed parameter; default to keeping what's there unless Ari has a reason to change it.
Rotation, scaling, and shear go in the appendix under "considered and rejected." Cheap to implement, but the coordinate frame makes them net-negative.

4.2 Temporal transforms

Candidates: speed jitter, temporal crop-resize, temporal flip, frame dropout.
Temporal speed jitter (resampling the clip by a random factor) is the single most consistently-cited skeleton-AR augmentation after spatial flip. It appears with various parameter ranges in the self-supervised literature (SkeletonCLR, AimCLR, A2MC), typically as "temporal crop-resize" with ratios in [0.5, 1.0]; in supervised long-tail work, Shap-Mix uses temporal segments of 40-70% of the original length. For mild in-distribution augmentation, the MDPI Electronics 2024 augmentation study and biomechanical intuition both suggest +-10-20% is the safe range: strokes do vary in duration by that margin across players in the professional pool, and the model should be invariant to that variation.
Implementation: pick a scale factor $s \in [0.85, 1.15]$ per clip; resample the F input frames to round($F \times s$) via linear interp on pose and court, nearest-neighbour on shuttle; then sample F frames at evenly-spaced positions from the resampled sequence (dropping from the edges when $s > 1$, padding edges when $s < 1$). **This operates within BST's existing clipped window, it does not change the clipping strategy**, which BST handles upstream (the "between two hits with max limits" clipping that produces the F-frame input, per data_pipeline_to_model_train.md). p=0.5 aligns with the literature norm for skeleton spatial/temporal augmentations.
Boundary with Phase 3. Phase 3 wants amateur-generalisation and will push the scale range much wider, plausibly [0.5, 1.5] or even [0.3, 2.0] to cover slower amateur strokes. The Week 2 arch-tuning doc had [0.7, 1.3] as a starting range; Phase 3 tunes up from there. **Task 2's range is deliberately narrower.** The principle: Task 2 samples the variation that already exists in pro-level ShuttleSet; Phase 3 synthesises out-of-distribution variation to bridge to amateur video. Same transform, different parameter, different intent. Both ranges will coexist in the final pipeline; the parameter needs to be configurable, not hardcoded.
Temporal flip (reversing the frame order) is a hard no. A reversed smash is not a smash, it's a preparation movement in reverse, which has no label in the taxonomy. Temporal flip is label-preserving on abstract motions (clap, wave) but not on stroke actions where the forward direction of motion is the entire discriminator. Also goes in the appendix under "rejected."
Frame dropout (randomly zeroing a small fraction of frames) is a cousin of joint masking and is plausible as a regulariser. Skip it for now: MMPose already produces dropouts in the real data at rates of 2-8% depending on class (per the hit-zone analysis), so the model is already being regularised by natural dropout. Adding synthetic dropout on top is double-counting. Flag as a possible follow-up once the natural-dropout rate stabilises post-Ari's MMPose re-extraction work.

4.3 Additive noise and joint masking

Candidates: Gaussian jitter on joint positions, joint masking, limb dropout.
PYSKL (Duan et al. ACM MM 2022, Towards Good Practices for Skeleton Action Recognition, arXiv:2205.09443) runs an explicit ablation on spatial augmentations for supervised skeleton AR on NTU-RGBD. Their finding: *"random rotation works for 3D skeletons; random scaling works for both 2D and 3D skeletons; while random Gaussian noise does not work for any kinds of skeletons."* This is the most direct piece of evidence for the supervised-classification regime, and it says Gaussian joint jitter is likely negative on clean skeleton data.
A small Gaussian jitter is recommended anyway, with one caveat. PYSKL was evaluated on NTU-RGBD, where skeletons come from Kinect depth sensors and are relatively clean. ShuttleSet skeletons come from MMPose on broadcast video, and have substantial per-frame jitter (3-10 pixels, per the Week 3 feature validation work). A Gaussian jitter augmentation that roughly matches the scale of the existing noise should act as a mild regulariser against over-reliance on exact joint positions, without adding noise beyond what the model already has to

handle. PYSKL's ablation is the stronger signal, and if Ari's run shows jitter doesn't help (or hurts), this augmentation is the first to drop.
Physical sanity check for σ = 0.005. At court-normalised scale, σ = 0.005 corresponds to ~6.7 cm of positional noise along the court's long axis (13.4 m) and ~3 cm along the short axis (6.1 m). MMPose's natural per-frame jitter on ShuttleSet is reported at 3-10 pixels; at typical broadcast resolution (~700 pixels per court length), that's ~6-19 cm of real positional noise. **σ = 0.005 sits at the low end of the real noise distribution**, we're matching (not exceeding) the noise the model already sees. Higher σ values (0.01-0.05 used in contrastive-learning literature, e.g. AimCLR at 0.01, SKELTER at 0.05) would exceed the natural noise distribution for this dataset and blur discriminative features between visually-similar strokes (passive drop vs drop, net shot vs cross-court net shot).
Joint masking (randomly zero-out a subset of joints per clip). The evidence base is mostly contrastive-learning (AS-CAL / Rao et al. 2021, AimCLR / Guo et al. 2022, the asymmetric augmentation work in Li et al. PMC 2022, which applies joint-mask at p=0.5 with 25% sub-probability activation inside a mix of augmentations). Mask-rate values in those works are typically 10-20% of joints with replacement; no clean supervised-only ablation establishes a best rate. Biomechanically, natural MMPose dropout is concentrated on specific joints (wrists during strikes, feet during jumps); uniform random masking at 10% teaches the model that any joint can go missing, which is a stronger regulariser than the natural dropout distribution alone. Parameter: mask 10% of the 34 joints (17 joints × 2 players) per clip, sampled uniformly. **p = 0.5**, aligning with the literature norm. Lower mask rates (5%) are safer; higher (20%) starts to destroy the action. Check what the BST code uses as its missing-joint sentinel (see Rule 6b).
Limb dropout (masking all joints of one limb simultaneously) is a structured version of joint masking. No strong case either way. It's cited in a couple of contrastive-learning papers but not in supervised benchmark results that I found. Appendix, for Ari's consideration if joint masking works.

4.4 Mixing augmentations (mixup, SkeletonMix, Shap-Mix)

This is the family where the long-tail literature is concentrated, and also the family where the multi-modal coupling problem is worst.
Mixup (Zhang et al. 2018), linear interpolation between two samples' inputs and labels, applied directly to skeleton data causes "severe unreasonable deformation to the skeleton topology" (Xu et al., Ta-CNN 2022). Two mid-air skeletons linearly interpolated produce a floating ghost that's neither of the source actions.
SkeletonMix (Xu et al. 2022, AAAI) addresses this by mixing upper-body joints from one sample with lower-body joints from another, preserving per-half topology. Works for single-subject datasets (NTU). *Does not work* on BST input: mixing upper body of a Smash with lower body of a Net shot produces a labelled chimera where the shuttle is still where the original Smash-sample put it, which is now inconsistent with the lower-body stance. The multi-modal coupling problem (Section 3) bites hard.
JMDA (Xiang and Wang 2024, ACM TOMM, "Joint Mixing Data Augmentation", DOI 10.1145/3700878) proposes SpatialMix + TemporalMix: spatial projection followed by mixing, and temporal-resize followed by mixing. Reports accuracy gains of up to ~2.8% on NTU-120 X-Sub with 2s-AGCN as a baseline, with more mixed results on NTU-60 (smaller gains and some slight regressions depending on the backbone). The temporal variant of JMDA has the same multi-modal inconsistency problem: mixing frame-61-90 of clip A with frame-61-90 of clip B leaves you with a blended stroke whose shuttle position is a linear combo of two different shuttle positions at the same frame index; that doesn't correspond to any physical event.
Shap-Mix (Zhang et al. IJCAI 2024, arXiv:2407.12312) is the most directly-relevant paper for our long-tail problem, it specifically targets long-tailed skeleton AR. Method: estimate joint

importance via Shapley values, then use a tail-aware mixing policy that preserves salient joints of minority classes. Reported improvements are substantial on long-tailed benchmarks they construct. **But it still has the multi-modal inconsistency problem**, and the Shapley estimation adds compute cost and training-pipeline complexity. It's also relatively new and not yet widely replicated.

Skip mixing augmentations for this document's scope. The multi-modal coupling violates Rule 1 of Section 3 for any spatial mix (pose A + pose B, with only one sample's shuttle) and violates Rule 4 for any temporal mix. The gains the papers report are real but were measured on single-modality skeleton data; whether they transfer to a multi-modal stack like BST is an open question and one that deserves its own dedicated experiment, not a bundled inclusion in a first-pass augmentation set.

Flag for later: if the four-augmentation recommendation lifts F1 meaningfully and Ari wants to push further, Shap-Mix (adapted to couple shuttle/court streams via the same sample-selection as the pose mix, i.e., don't mix the shuttle, just use the Shap-selected sample's shuttle as-is) is the most promising direction. That's a standalone piece of work.

4.5 Long-tail sampling (distinct family, lower depth)

Augmentation grows the effective training signal uniformly. Class-weighted CE and label smoothing (both already in place) bias the gradient updates. Neither directly addresses the "the model sees a Rush only once every 100 minibatches" problem, which is where **sampling-side interventions** live.

WeightedRandomSampler (PyTorch) replaces the default shuffled sampler with a weighted one, typically weighted ∝ 1/class_count to produce a class-balanced minibatch distribution. This is the sampling-side analogue of class-weighted loss, and is straightforward to swap in: set per-class weights, feed to the sampler, DataLoader handles the rest.

he key empirical finding from the long-tail literature: **Kang et al.'s "Decoupling Representation and Classifier for Long-Tailed Recognition" (ICLR 2020, arXiv:1910.09217)**. On image classification benchmarks, instance-balanced sampling (the default, i.e., each sample equally likely regardless of class) is actually *better* for representation learning than class-balanced sampling, and class-balanced sampling is better for the classifier head. Their two-stage training (cRT and LWS variants) is a fix. Whether this translates to skeleton transformers is not directly established, but the principle (uniform class-balanced sampling can hurt feature learning) should inform sampler choice.

For this project, three reasonable sampling strategies:

- **Uniform (no change).** Each clip in the train set equally likely. Current behaviour.
- **Inverse-frequency (1/count).** Each class equally likely per minibatch. Standard WeightedRandomSampler recipe. Over-amplifies the long tail; risks over-fitting to the ~350 Rush clips if Rush is sampled 30× its natural rate.
- **Square-root frequency (1/√count).** A middle ground widely used in the long-tail literature as a compromise between uniform and class-balanced. It amplifies the tail without collapsing to one-clip-per-class-per-batch. For our 12× imbalance (Rush ~470 to Net shot ~5,800), √-weighting gives rare-class amplification of ~3.5×, which is meaningful but not extreme.

Recommendation: use WeightedRandomSampler with weights ∝ 1/√(class_count). Applied only during training. Validation and test remain instance-balanced (which is what you want for reporting per-class F1 honestly).

This is a distinct family from augmentation proper. It doesn't generate new samples, it changes the rate at which existing samples are seen.

Out of scope: focal loss, logit adjustment, class-balanced loss (Cui et al. 2019), balanced softmax (Ren et al. 2020), Kang et al.'s two-stage cRT / LWS recipes. These are **loss-side**

engineering interventions and are adjacent to but not the same as sampling-side. They deserve their own separate investigation.

5. Recommendation

5.1 The augmentation set

#	Augmentation	Parameters	p	Streams	Literature anchor
1	**Centreline flip** (mirror across court centreline, x -> W - x)	Coordinate mirror + bilateral joint-index swap	0.5	pose + shuttle + court, coupled	Shap-Mix (Zhang et al. IJCAI 2024); standard across skeleton-CL lit
2	**Temporal speed jitter**	scale ∈ [0.85, 1.15], linear interp on pose/court, nearest-neighbour on shuttle, resample within existing BST clip window (no change to clipping strategy)	0.5	pose + shuttle + court, coupled, same scale	MDPI Electronics 13(4):747 2024 augmentation study; Shap-Mix uses 40-70% for extreme, +-15% is the mild end
3	**Gaussian joint jitter**	σ = 0.005 in court-normalised units, per-joint i.i.d.	0.5	pose only	AimCLR (Guo et al. AAAI 2022, arXiv:2112.03590) uses N(0, 0.01) for contrastive; SKELTER σ=0.05 for robustness; ours scaled down to match natural MMPose noise. **PYSKL ablation found Gaussian noise net-negative on clean skeletons, see Section 4.3**
4	**Random joint masking**	Mask 10% of 34 joints per clip (uniform draw), replace with sentinel matching existing missing-joint convention	0.5	pose only	Rao et al. 2021; Liu et al. 2025 survey Section 4

5.2 Sampling

Change	Value
WeightedRandomSampler weights	w_c ∝ 1 / sqrt(class_count_c) where c is the class label
Applied at	Training DataLoader only; val and test remain instance-balanced

5.3 Ordering and composition

The four augmentations are independent and order-matters-a-little. Recommended order per clip:

1. **Flip** (applied first, clean spatial transform)
2. **Temporal speed jitter** (applied next, clean temporal transform)
3. **Gaussian joint jitter** (additive noise, applied after spatial/temporal so it's the last thing the model sees)
4. **Joint masking** (applied last, explicit "missing" signal)

Reasoning: flip and temporal jitter are deterministic transforms of clean data; applying noise and masking before them means the noise / mask gets transformed too, which makes the effective σ and mask rate drift off their configured values.

5.4 What this should look like in __getitem__

In pseudo-code:

```
def __getitem__(self, idx):
    clip = self.load_clip(idx)     # dict with keys: pose, shuttle, court, label,
hit_frame
    clip = random_flip(clip, p=0.5)                             # Aug 1; flips all 3
streams, swaps joint indices
    clip = random_temporal_speed(clip, scale_range=(0.85, 1.15), p=0.5)  # Aug 2;
coupled resample
    clip["pose"] = gaussian_jitter(clip["pose"], sigma=0.005, p=0.5)     # Aug 3;
pose only
    clip["pose"] = random_joint_mask(clip["pose"], mask_frac=0.10, p=0.5) # Aug 4;
pose only
    return clip
```

Plus, once at DataLoader-construction time, replace the default RandomSampler with WeightedRandomSampler(weights=1/sqrt(class_count), num_samples=len(train_set), replacement=True).

6. Reconciliation with the existing pipeline

First-order finding: BST's training loop applies exactly one augmentation, and it violates the multi-modal coupling rule from Section 3.

RandomTranslation_batch (shuttleset_dataset.py:123) is instantiated at bst_train.py:243 and applied at lines 93 (human_pose = random_shift_fn(human_pose)) and 98 (joints = random_shift_fn(joints)). Parameters: trans_range=(-0.3, 0.3), prob=0.3. Joints-only, batch-level. Docstring identifies it as ported verbatim from TemPose.

The dataset-level RandomTranslation class (shuttleset_dataset.py:108) is present but not live:

- Dataset_npy.__getitem__ calls self.random_shift(joints) at line 202 and discards the return. Because x = x + shift inside the class rebinds a local name rather than mutating the input, the joints array is unchanged. Silent no-op.
- Dataset_npy_collated docstring at line 240 explicitly states "Notice: There is no random translation here." Three further collated variants (lines 520, 554, 588) carry the same disabled-docstring.
- historical_predecessor_analysis_summary.md:144 already documents this situation. Someone on the team found the same thing independently.

So far we have no augmentation: one joints-only xy-shift, inherited not authored, applied at the batch level during training. Nothing on shuttle. Nothing on court. Nothing temporal. Nothing pose-internal.

Two consequences for the Section 5 recommendations:

1. The existing RandomTranslation_batch violates Rule 1 of Section 3, because pose is translated and shuttle and court are not. The multi-modal coupling problem Section 3 describes is already present in the codebase, not hypothetical.
2. The +-0.3 range is aggressive relative to the recommended parameters; the absolute magnitude needs confirming against the joint-coordinate normalisation scheme. Open question for Ari, keep it alongside the new augmentations, tune it, or replace it with a coupled court-level translation applied identically to all three streams.

Open items for Ari

Question	Why it matters
Exact line(s) of code where RandomTranslation is applied	Determines whether it's pose-only, or also applied to shuttle/court streams
Parameter values (translation range, probability)	Whether it's a no-op (range=0), a reasonable mild aug (range~0.02), or something more aggressive
Whether it's applied in training only or also val/test	Val/test application would be a bug, translation shouldn't be at eval time
Whether the shuttle and court-position streams are translated by the same offset	If not, this is already violating Rule 1 of Section 3, and the existing aug is actively mis-training the cross-attention

Reconciliation matrix (my recommendation vs. the existing code, pending confirmation)

Source	States	My recommendation	Status
BST paper (Chang 2025)	No augmentation reported	4 augmentations + weighted sampler	**Substantially accurate.** BST code inherits a single joints-only xy-shift from TemPose verbatim. No

Source	States	My recommendation	Status
			authored augmentation. Paper's 'no augmentation' claim essentially holds.
Existing dataloader	RandomTranslation present (details TBC)	Keep, tune, or replace depending on parameters	RandomTranslation_batch joints-only at bst_train.py:93, 98 (+-0.3, p=0.3). Dataset-level class is dead code" and change Status to "Violates Rule 1 of Section 3 as-is. Recommendation: replace or extend with a coupled translation applied identically to pose, shuttle, and court."
Class-weighted CE + label smoothing 0.1	Already in training loop	Keep; add WeightedRandomSampler on top	**Composes.** Class weights bias gradients; sampler biases exposure rate. Using both is standard for long-tail (Kang et al. 2020 Section 4.2 discusses this interaction, noting diminishing returns but no outright conflict)
Arch doc Phase 3 plans	Temporal speed + camera aug for amateur generalisation	Narrower speed range here, wider in Phase 3	**Complements.** Explicit boundary in Section 7
Arch 1 doc (X3D-S wrist crop)	RGB crop augmentation territory	Out of scope	**Separate concern.** Flagged as potential Task 2b in Section 7
tuning_thoughts.md	Label smoothing, class-weighted CE	Keep as-is	**Aligns.**
Task 1 writeup	Filter pose stream for X3D-S consumer	Independent of this work	**No overlap.** Task 1 operates on MMPose output before collation; Task 2 operates on collated tensors inside the DataLoader

What changes in the dataloader

Three things:

5. Replace / extend RandomTranslation (depending on what it is, see above)
6. Add the four new augmentations, composed in the order from Section 5.3
7. Swap the sampler for WeightedRandomSampler with √-frequency weights

Implementation note: the coupling helpers (a single random_flip(clip_dict) that handles all three streams with correct joint-index swapping) should be written once and reused. The fiddly parts are (a) the joint-index swap table for the horizontal flip, (b) the evenly-spaced resample in temporal speed jitter, which needs the same index mapping applied across pose/court/shuttle, (c) making sure the shuttle stream's nearest-neighbour resample doesn't drop known hit frames (if any are tracked in metadata).

7. Caveats

8. **Parameter values are anchored, not empirically tuned on ShuttleSet.** Rotation-free design, flip p=0.5, temporal scale ++-15%, jitter σ=0.005, mask 10%, √-frequency sampling. Each is a starting point with a literature reference, and each wants verification on actual ShuttleSet training runs."
9. **Phase 3 (amateur generalisation) is a separate and wider-ranging work.** The temporal speed range in this writeup is +-15%, that's in-distribution pro-to-pro variation. Phase 3 will need something like +-30-50% to cover amateur slow strokes. The *parameter* changes, not the *transform.* Implementation implication: the speed-jitter scale range needs to be configurable in the dataloader, not hardcoded, so Phase 3 can extend it.
10. **Arch 1 X3D-S wrist-crop RGB augmentation is out of scope.** Ari's Arch 1 doc gestures at ColorJitter, random erasing, RandAugment for the 112×112 wrist crops. That's a completely different regime: different literature (image classification / video action recognition, not skeleton AR), different parameter space (pixel-level vs. joint-level), and a different multi-modal concern (the crop's anchoring to the wrist coordinate means crop augmentation interacts with pose augmentation). **Flag as Task 2b** if Ari wants it covered. Not combining. The two writeups have different sources and different failure modes.
11. **Mixing augmentations skipped, with reason.** Shap-Mix and JMDA are the specifically-long-tail-designed skeleton-AR methods in the recent literature. They don't make the cut for this pass becausec the multi-modal coupling problem (Section 3 Rule 1) makes the gap between their reported single-modality gains and what we'd get on BST's three-stream stack hard to predict without a dedicated experiment. Flagged as a follow-up direction if the four-augmentation set leaves F1 on the table.
12. **Sampling can over-amplify rare classes.** 1/sqrt(count) is the safer choice than 1/count for exactly this reason. With Rush at ~350 clips, 1/count sampling means the model sees the same ~350 clips many times per epoch, which increases overfitting risk. √-weighting makes the tradeoff gentler. If Ari wants to go more aggressive, 1/count is the other obvious point to try; anything between is fine.
13. **No evidence yet that skeleton augmentation specifically helps the *target* classes** (Smash, Wrist smash, Long service, Rush). The literature evidence is that augmentation helps average performance on balanced benchmarks (NTU); the long-tail-specific literature (Shap-Mix, Balanced Representation Learning) shows it helps rare classes in general, but those benchmarks are synthetic long-tail versions of NTU, not this dataset.
14. **"Number of joints" is 17 × 2 players = 34, not 17.** Joint masking at 10% means ~3 joints masked per clip, which might be one or both players' wrist, one player's ankle, etc. Masking needs to be sampled over the full 34-joint set, not per-player with a 10% rate. The BST code treats the two-player dimension distinctly; masking code needs to match.

8. Appendix

8.1 Considered and rejected

Technique	Reason
Rotation (+-15-30° around court centre)	Court-relative coordinates already normalised to canonical frame; rotating moves feet off the baseline, breaks court geometry for cross-attention. Net-negative on this input representation
Scaling (×[0.9, 1.1])	"Player size" in court coords is a proxy for projection quality; randomising it teaches the model that broken homographies are normal
Shearing (+-0.1)	Deforms court geometry; shuttle trajectory starts looking non-physical
Mirror across the NET (top-bottom flip, y -> H - y)	Requires slot-swap + label-consistency check; flips shuttle direction relative to the striking player. Feasible in principle but fiddly and not standard in the skeleton-AR literature. Revisit if the centreline-only flip leaves obvious symmetry on the table
Temporal reverse	A reversed smash has no valid label in the taxonomy. Trivially not label-preserving for stroke actions
Synthetic frame dropout	MMPose already produces natural dropout at 2-8% per class. Synthetic on top is double-counting; revisit if natural-dropout rate changes post-Ari's re-extraction
Vanilla mixup	Produces floating-ghost interpolations; well-documented failure mode in skeleton literature (Xu et al. 2022)
SkeletonMix (upper/lower body swap)	Multi-modal coupling, swapping one sample's upper body with another's lower body while keeping one sample's shuttle and court positions produces physically inconsistent clips
JMDA SpatialMix / TemporalMix	Same multi-modal issue as SkeletonMix; papered-over on NTU because NTU is single-modality
Shap-Mix	Most relevant long-tail method, but (a) multi-modal coupling unclear, (b) implementation cost non-trivial, (c) replicability of published gains still being established. Flag as follow-up if Section 5 set makes F1 viable.
Focal loss / logit adjustment / class-balanced loss / balanced softmax	Loss-side engineering, out of scope per the framing call. Separate investigation if needed
AutoAugment / RandAugment / learned augmentation	Requires augmentation-search budget we don't have and data volumes larger than ShuttleSet to search over. Appendix mention only

Technique	Reason
Limb dropout (all joints of one limb masked together)	No strong evidence either way; structured variant of joint masking. Possible extension if Section 5 joint masking proves helpful

8.2 Open questions for Ari

15. What exactly does RandomTranslation do in the current dataloader? Parameters, streams covered, train-only vs train+eval?
16. Does the pose_style flag (J, J+B, J2B, J+2B) affect how horizontal flip should compose? Bone vectors transform identically to joint positions under mirror-x, but the joint-index swap table is the same either way. Just confirming this is as simple as I think.
17. Missing-joint sentinel convention, is it (0, 0) or a specific value (NaN, -1, etc.)? Determines the mask-replacement value for Aug 4.
18. Does the clip metadata carry any frame-index markers (hit frame, strike frame) alongside the F-frame tensor? Temporal speed jitter as recommended doesn't require them, it just evenly-samples F frames from the resampled sequence, but if they exist, they must be updated under the resample mapping, and it opens up a hit-frame-preserving resample as a variant worth comparing if the evenly-spaced version struggles on short strokes.

8.3 Papers worth having on file (for the final report's related-work section)

- Chang, J.-Y. *BST: Badminton Stroke-type Transformer*. arXiv:2502.21085, 2025. Our backbone.
- Ibh, Grasshof, Witzner, Madeleine. *TemPose: A New Skeleton-Based Transformer*. CVPR Workshops 2023. BST's lineage.
- Duan, Wang, Chen, Lin. *PYSKL: Towards Good Practices for Skeleton Action Recognition*. ACM MM 2022, arXiv:2205.09443. The closest thing to a supervised-AR augmentation ablation in the literature. Finds Gaussian noise net-negative on clean skeletons. Directly informs Section 4.3 caveat.
- *Enhancing Human Action Recognition with 3D Skeleton Data: A Comprehensive Study of Deep Learning and Data Augmentation*. MDPI Electronics 13(4):747, 2024, doi:10.3390/electronics13040747. Augmentation taxonomy reference.
- Liu, Liu, Hu, Ren, Yuan, Lin, Wen. *3D Skeleton-Based Action Recognition: A Review*. arXiv:2506.00915, 2025. More recent and more complete survey.
- Zhang et al. *Shap-Mix: Shapley Value Guided Mixing for Long-Tailed Skeleton Based Action Recognition*. IJCAI 2024, arXiv:2407.12312. Directly long-tail, skeleton-specific, flagged as follow-up.
- Liu et al. *Balanced Representation Learning for Long-tailed Skeleton-based Action Recognition*. Machine Intelligence Research 2024, arXiv:2308.14024. Long-tail methods reference.
- Kang, Xie, Rohrbach, Yan, Gordo, Feng, Kalantidis. *Decoupling Representation and Classifier for Long-Tailed Recognition*. ICLR 2020, arXiv:1910.09217. Sampling-strategy evidence.
- Rao et al. *Augmented Skeleton Based Contrastive Action Learning (AS-CAL)*. Information Sciences 2021. Standard contrastive augmentation parameters.
- Xiang et al. *Joint Mixing Data Augmentation (JMDA)*. ACM TOMM 2024. Mixing on NTU, rejected here but the FeatureAlignment insight is a useful template for future multi-modal mixing work.

- Xu et al. *Topology-aware CNN / SkeletonMix*. AAAI 2022, arXiv:2112.04178. The "mixup destroys skeleton topology" finding.
- Cormier, Schmid, Beyerer. *Enhancing Skeleton-Based Action Recognition in Real-World Scenarios Through Realistic Data Augmentation*. WACV Workshops 2024. "Skelbumentations" library; most directly targeted at realistic pose-estimation-noise augmentation. Useful reference implementation.

9. Summary

Four augmentations for the BST dataloader: horizontal flip (p=0.5), temporal speed jitter +-15% (p=0.5), Gaussian joint jitter σ=0.005 (p=0.5), random joint masking 10% (p=0.5). Plus WeightedRandomSampler with √-frequency weights for the long-tail push. Parameters are anchored, flagged for your ablation. PYSKL's supervised ablation found Gaussian noise net-negative on clean skeletons, so jitter is the first to drop if ablation shows it doesn't help

On RandomTranslation: one augmentation is live, a joints-only ±0.3 xy-shift in the training loop (bst_train.py:93, 98), ported verbatim from TemPose.. The dataset-level class is dead code (silent return-discard bug at Dataset_npy line 202, explicitly disabled in Dataset_npy_collated). Your framing holds. What exists is minimal, inherited, and violates Rule 1 of Section 3 because shuttle and court aren't co-translated. First thing in the reconciliation table is "what does RandomTranslation currently do and should it stay."

Explicit scope boundaries: no amateur-generalisation work (that's Phase 3, wider parameters), no X3D-S crop augmentation (different regime, flagged as possible Task 2b). Mixing augmentations (Shap-Mix, JMDA) are out: the multi-modal coupling problem on BST is worse than on NTU.